

CSS

Transition, float, media query

Theory

1. Float Property

- **Definition:** Positions an element to the left or right of its container, allowing text/content to wrap around it.
- **Common use cases:** Image alignment, simple layouts (though modern CSS prefers Flexbox/Grid).
- **Drawback:** Can cause layout issues if not cleared properly (clear: both).

2. Flexbox

- **Definition:** One-dimensional layout model for arranging items in rows or columns.
- **Key properties:**
 - display: flex → activates flexbox.
 - justify-content → horizontal alignment.
 - align-items → vertical alignment.
 - gap → spacing between items.

3. Media Queries

- **Definition:** CSS technique for responsive design.
- **Syntax:**

```
@media (max-width: 576px) {  
  div { flex-direction: column; }  
}
```

- **Use case:** Adjust layout for different screen sizes (mobile, tablet, desktop).

4. CSS Transitions

- **Definition:** Smoothly animates property changes.
- **Properties:**
 - transition-property → which property to animate.
 - transition-duration → how long.

- transition-timing-function → speed curve (ease, linear, ease-in-out).
- transition-delay → wait before starting.
- **Shorthand:** transition: all 2s ease-in;

Coding Round Questions

- Float → Explain float with an example of image alignment.
- Flexbox → Create a flex container with items centered.
- Media queries → Write a media query for mobile layout.
- Transitions → Show how to animate background color on hover.
- Responsive design → Explain how media queries make layouts responsive.
- Flex vs Float → Compare float and flexbox for layout design.

⚡ Machine Coding Round Challenge

Problem: Responsive Card Layout

Requirements:

- Create a container with two cards side by side using Flexbox.
- On mobile (max-width: 576px), stack cards vertically.
- Add hover effect with background color change and smooth transition.
- Use float to align an image inside a card.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Float</title>
    <style>
      /* div{
        border: 2px solid black;
        height: 400px;
        float: right;
      }

      div img {
        float: right;
      } */
      /*
      div {
        background-color: yellow;
        width: 300px;
```

```

    height: 300px;
    text-align: center;
} */

/* from 0 to 540px */
/* @media (max-width: 540px) {
    div {
        background-color: purple;
    }
} */

/* from 541 px to .... */
/* @media (min-width: 541px) {
    div {
        background-color: orange;
    }
} */

/* @media (min-width: 720px) and (max-width: 800px) {
    div {
        font-size: 60px;
    }
} */

#container {
    display: flex;
    border: 2px solid black;
    gap: 20px;
    height: 300px;
    padding: 10px;
}

#container div {
    border: 2px solid black;
    width: 45%;
    /* transition-property: background-color , font-size; */
    /* transition-property: all; */
    /* transition-duration: 4s; */
    /* transition-delay: 5s; */
    /* transition-timing-function: ease-in-out; */
    /* shorthand property */
    transition: all 2s ease-in ;

    /* border: 20px solid yellow; */
}

#container div:hover {

```

```

        background-color: aqua;
        font-size: 10px;
    }

    @media (max-width: 576px) {
        #container {
            flex-direction: column;
        }
        #container div {
            width: 100%;
        }
    }
</style>
</head>
<body>
    <!-- <div>
        
    </div>

    <div>
        
    </div> -->

    <div id="container">
        <div class="left"><h1>left</h1></div>
        <div class="right"><h1>Right</h1></div>
    </div>
</body>
</html>

```